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Dublin City University

SEMESTER 1 EXAMINATIONS 2024/2025

MODULE: CSC1042 - Computer Networks 2

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

YEAR OF STUDY: 1,3

EXAMINER(S):

Brian Stone	(Internal)	(Ext:)
Mohammed Amine Togou	(Internal)	(Ext: 8929)
Michael Scriney	(Internal)	(Ext:)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer all questions.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1 ***[TOTAL MARKS: 30]***

Q 1(a) **[20 Marks]**

The IP address 192.168.100.0 is to be subnetted. Using 3 bits for subnetting, answer the following questions:

1. What class is this address? Explain how you find it.
2. What is the default subnet mask of this class?
3. What is the new subnet mask of this address?
4. How many subnets can be created?
5. What are the IP addresses of these subnets?
6. What is the broadcast address of each subnet?
7. What are the first and last host IP addresses of each subnet?

Show all you work clearly and use full 4 bytes decimal-dot-notation throughout.

Q 1(b) **[10 Marks]**

Given the network address 139.10.0.0/14, answer the following:

1. How many Class B networks are included in this supernet?
2. What are their IP addresses?
3. What is the broadcast address of this supernet?
4. What is the first and last valid host IP addresses in this supernet?

[End of Question 1]

QUESTION 2 ***[TOTAL MARKS: 28]***

Q 2(a) **[6 Marks]**

Abbreviate the following IPv6 address:

00ab:7500:0000:f2b0:bacf:0000:0000:fa5d

Q 2(b) **[10 Marks]**

Write the lines to be added to the SAO record to create the following:

1. Type A record that maps the domain name “books.com” to the IP address 205.17.10.1
2. Type CNAME record for “drama” that points to “books.com”
3. Type NS record for “ns1”, the nameserver responsible for resolving DNS queries of “books.com”
4. Type PTR record to enable reverse DNS lookup for “books.com”

Q 2(c) **[8 Marks]**

The following firewall rules are to be implemented:

1. Allow established TCP connections to all IP addresses
2. Forbid all connections except those on this list
3. Allow access to the DHCP server only from within the network
4. Allow access to Apache web server on port 80

In what order would you place these rules to achieve an efficient drop policy.

Q 2(d) **[4 Marks]**

Mike, an IT intern at ACME Inc., noticed some strange packets while revising the security logs generated by the firewall. A handful of IP addresses on the internet were sending malformed packets to several different IP addresses, at several different random port numbers inside ACME Inc. What type of attack is described in this scenario?

[End of Question 2]

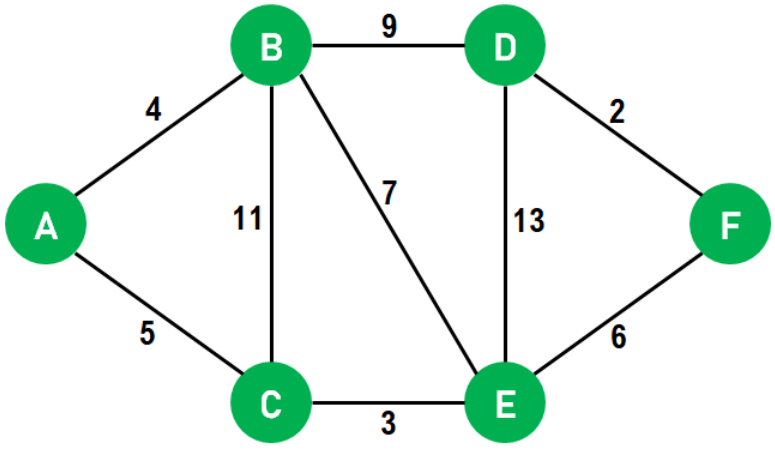
QUESTION 3 ***[TOTAL MARKS: 42]***

Q 3(a) **[10 Marks]**

Explain how a router creates its routing table using a link state routing approach.

Q 3(b) **[8 Marks]**

Using the graph below and following the Dijkstra's algorithm, show the order through which the nodes will be visited to compute shortest path from node A to node F.

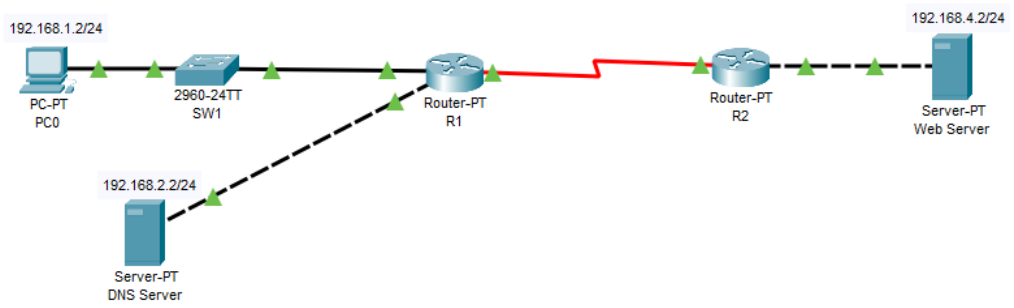


Q 3(c) **[12 Marks]**

Given the network depicted in the figure below, answer the following:

1. How many networks are there?

2. What should be configured at the level of the routers to enable communication between these networks?
3. Assume the Web server has the HTML code for www.example.com and that the DNS server has the record that maps www.example.com to the IP address 192.168.4.2. Assume that ARP has been used to get the MAC addresses of all devices. Describes what happens when a user opens the web browser on PC0, types in www.example.com and hits enter. Indicate the order of protocols used.



Q 3(d) **[6 Marks]**

What are the steps required to troubleshoot a computer/network problem following the troubleshooting methodology?

Q 3(e) **[6 Marks]**

Give two security measures that are used to mitigate device and network attacks.

[End of Question 3]

[END OF EXAM]